HTE staff checklist - Setup

Edition 2026-10-11 | Local OT-2 | 66 first-round + 10 second-round wells

Campaign folder: _____ Operator/date: _____

Before use: HTE must release the block/seal/mount, pipetting heights, extraction/IS recovery, LC method/matrix and a measured plan within 12 h. These remain pending; live export enforces LC settings and timing. Use released files.

Before the clock

- Have the real batch/stock/deck CSVs, LC sequences and Python protocols. Check **65 pairs + one L17 reference; all 31 ligands covered in pairs**. Start the local export hook/watcher; key file: `anthropic.key` on the laptop.
- Prepare stocks from `stock_preparation.csv`, including reserves and assay corrections. Prepare ligands in the glovebox. Label IDs/addresses; dispense only homogeneous stocks; cap reserves.

Stock	Preparation / dispensing concentration
SM + naphthalene	250 mM SM + 2.5 mM IS in DCM; prepare at least 5.00 mL.
All 31 ligands	30 mM in toluene; CSV gives each mass and reserve.
Pd(COD)(DQ)	60 mM in toluene; verify identity/assay and solubility.
Toluene	Neat.
Workup / LC diluent	Separate sources: 1:1 water/ACN + 1% v/v HCOOH; no IS.

SM: pyridine-2-sulfonyl fluoride (PyFluor). **Product:** 2-fluoropyridine. Combined-stock recipe at 100% assay: 201.45 mg SM + 250 uL of 50 mM naphthalene/DCM master; DCM to final 5.00 mL in a capped 20 mL glass/PTFE vial. Naphthalene master: 12.817 mg to final 2.00 mL; prepare separate DCM and LC-compatible masters.

OT-2 deck - load stage quantities from `deck_loads.csv`

Left: p20 single GEN2. **Right:** p300 eight-channel GEN2. Manual block moves only.

Slot	Contents
1 / 4	Ligand racks: L01-L24 / L25-L31, at the exact CSV well addresses.
2	12-channel reservoir: A1 SM/IS/DCM; A2 Pd; A3 toluene; A4 workup; A5 LC diluent.
3 / 10	Reaction glass-vial block for dosing/cooling / heater-shaker with released adapter.
6	Second-round substrate array during first dosing; LC plate during workup.
5 / 8 / 9	p20 tips / p300 tips / p300 tips. Keep slots 7 and 11 empty.

LC calibration - prepare stocks now; run standards during round one

Authentic SM/product stocks: **100 mM, 2.00 mL each, no IS** (SM 32.232 mg; product 19.418 mg at 100% assay; correct from COA). Working analyte: 100 uL of 100 mM to 1.00 mL (10 mM). Analytical IS: 40 uL of LC-compatible 50 mM master to 2.00 mL (1 mM).

For **each analyte**, make 1.00 mL per concentration column, with **50 uL of 1 mM IS in every tube**. Use the approved matrix recipe, accounting for stock solvents.

Analyte, uM	0 blank	100	500	1,000	2,500	6,000	5,000 check
Stock, mM	None	10	10	10	100	100	100
Analyte, uL	0	10	50	100	25	60	50
Matrix solvent, uL	950	940	900	850	925	890	900

Fit the blank and five nonzero levels; keep the independent 5,000 uM check outside the fit. **14 injections total**, during the first reaction.

HTE staff checklist - Run

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Block/seal: _____ LC method/channel: _____

Follow the local App's pauses and the released CSV well maps. Keep plate orientation and sample IDs intact. Start the competition timer with first-round dosing; stocks and first-batch selection are prepared beforehand.

Step	Staff action
1. First dosing	Import `round1/dose_and_react.py` into the local OT-2 App. Confirm offsets/deck. Dose **20 uL combined SM/IS/DCM into 66 wells** in slot 3.
2. Remove DCM	Manually move the open block to slot 10. Use the released dry-down procedure (initial proposal: 37 C, 300 rpm, 10 min; requires HTE validation). Verify the dry endpoint; return to slot 3 as prompted.
3. Assemble and react	Per pair: **10 uL ligand A + 10 uL ligand B + 20 uL toluene + 10 uL Pd stock**. Reference: **20 uL L17 + 20 uL toluene + 10 uL Pd**. Seal, move to slot 10 and run **95 C / 500 rpm / 4 h**, using the validated thermal-lag procedure.
4. While reacting	At the App pause, predose **20 uL SM/IS/DCM into ten second-round wells** using `round2_predose.csv`. Remove that array to the approved ventilated ambient-evaporation location; resume promptly. Run the **14 calibration injections** in parallel. Cap/remove volatile sources at the permitted pauses; remove covers before any aspiration.
5. Cool	At the end of four hours, follow the manual-move prompt and thermal-handling SOP. Move the sealed block off the heater to slot 3. Cool before opening.
6. Work up	Load LC plate in slot 6 and run `round1/workup.py`. Add **50 uL workup solvent, without additional IS**, then shake the reaction block **2 min without heating** using the released method. Return to slot 3.
7. Dilute and analyze	Check homogeneous extract; remix if delayed and let bubbles clear. Fresh tips prewet **twice in their own reaction well** before slow aspiration. Robot adds **180 uL diluent + 20 uL extract** to mapped LC wells; discard tip surplus. Seal, mix LC plate **2 min**, let bubbles clear; check needle height/volume. Run **66 samples** using `lc_sequence.csv`.
8. Trigger next batch	Export the complete integrated peak list and publish its completion marker through the configured local hook. Wait for QC to pass and for `next_design.csv` / `next_dosing.csv` (**ten new pairs**). Follow the configured round-two export to obtain the released local protocols and deck-load CSV. If QC or export stops, resolve it before dosing.
9. Second reaction	Check the ten predosed wells and orientation; **verify DCM is gone**. Load stage quantities from the new deck CSV. Run the released second-round dosing protocol: add the selected ligands, toluene and Pd as in step 3; **do not redose SM/IS or perform a second heater dry-down**. Seal and react at **95 C / 500 rpm / 4 h**.
10. Finish	Cool sealed, then repeat steps 6-7 with the second-round workup protocol for **ten samples**. Stop the clock when all ten analyses are complete. Save calibrated results, QC, stock/refill records and Opentrons JSON logs.

Pause and correct before resuming

- Bottom contact, unintended air, dripping, bubbles or solids: pause and correct. Validate clearance/immersion (sample starts 1 mm above bottom); do not force insoluble material through tips.
- Reuse only same-stock dosing tips with validated noncontact dispensing. Sample tips stay fresh. Never return a tip to source after LC contact. An uncertain transfer needs a fresh documented dilution, never a guessed top-up.
- Two phases/solids: stop sampling and use the approved extraction method. Keep reactions sealed through heating/cooling. Preserve failed measurements; repeat only for documented QC faults, never to choose a higher yield.

Per reaction: 5 umol SM, 50 uL final toluene solution, 0.1 M; 6 mol% each ligand, 12 mol% Pd; no base or external fluoride. **Timing:** 2 min per LC sample; 76 reaction samples total. The provisional 729-min plan must be shortened and measured before release against the 720-min limit.

Detailed procedures and unresolved settings remain in the reference handbook and HTE configuration checklist.